

# HDOC Day-23

## C Programming Assignment — Basic Square / Rectangular Patterns

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|----------------|---------------|
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| <b>Program</b> | B.Tech CSE    |

### Objective

Write a menu-driven C program using a switch statement to print the selected basic square/rectangular pattern. Use nested loops, without arrays. Each pattern is implemented as a separate parameterless function with no return value.

### Algorithm

1. Start the program.
2. Display the menu containing the four pattern choices.
3. Read the user's choice.
4. Read the required number of rows and/or columns.
5. Use a switch statement to call the corresponding pattern function.
6. Use nested loops inside the selected function to print the pattern.
7. Display the output and stop.

## C Program

The following single program provides all four activities through a switch-based menu.

```
#include <stdio.h>

int rows, cols;

void solidSquare()
{
    for (int i = 1; i <= rows; i++)
    {
        for (int j = 1; j <= rows; j++)
            printf("* ");
        printf("\n");
    }
}

void solidRectangle()
{
    for (int i = 1; i <= rows; i++)
    {
        for (int j = 1; j <= cols; j++)
            printf("* ");
        printf("\n");
    }
}

void rowNumber()
{
    for (int i = 1; i <= rows; i++)
    {
        for (int j = 1; j <= cols; j++)
            printf("%d ", i);
        printf("\n");
    }
}

void columnNumber()
{
    for (int i = 1; i <= rows; i++)
    {
        for (int j = 1; j <= cols; j++)
            printf("%d ", j);
        printf("\n");
    }
}

int main()
{
    int choice;

    printf("HDOC Day-23\n");
    printf("1. Solid square pattern\n");
    printf("2. Solid rectangular pattern\n");
    printf("3. Row-number pattern\n");
    printf("4. Column-number pattern\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("Enter number of rows: ");
            scanf("%d", &rows);
            solidSquare();
            break;

        case 2:
        case 3:
        case 4:
            printf("Enter number of rows: ");
```

```
scanf("%d", &rows);
printf("Enter number of columns: ");
scanf("%d", &cols);

switch (choice)
{
    case 2: solidRectangle(); break;
    case 3: rowNumber(); break;
    case 4: columnNumber(); break;
}
break;

default:
    printf("Invalid choice.\n");
}

return 0;
}
```

## 1. Solid square pattern

**Sample input:**

Enter number of rows: 4

**Output:**

```
* * * *
* * * *
* * * *
* * * *
```

## 2. Solid rectangular pattern

**Sample input:**

Enter number of rows: 3

Enter number of columns: 5

**Output:**

```
* * * * *
* * * * *
* * * * *
```

## 3. Row-number pattern

**Sample input:**

Enter number of rows: 4

Enter number of columns: 4

**Output:**

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

## 4. Column-number pattern

**Sample input:**

Enter number of rows: 4

Enter number of columns: 4

**Output:**

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

## Test-Case Table

| TC | Choice | Input                 | Expected output / result                    |
|----|--------|-----------------------|---------------------------------------------|
| 1  | 1      | Rows = 4              | 4 × 4 solid square of *                     |
| 2  | 2      | Rows = 3, Columns = 5 | 3 × 5 solid rectangle of *                  |
| 3  | 3      | Rows = 4, Columns = 4 | Rows contain 1s, 2s, 3s and 4s respectively |
| 4  | 4      | Rows = 4, Columns = 4 | Each row prints 1 2 3 4                     |
| 5  | 9      | —                     | Invalid choice message                      |

## Conclusion

The menu-driven C program successfully uses switch statements and nested loops to generate the required square, rectangular, row-number, and column-number patterns without using arrays.

**Program:** B.Tech CSE    **Batch:** 2026–2030